NotebookLM Clone

Final Project Evidence and Learning Overview

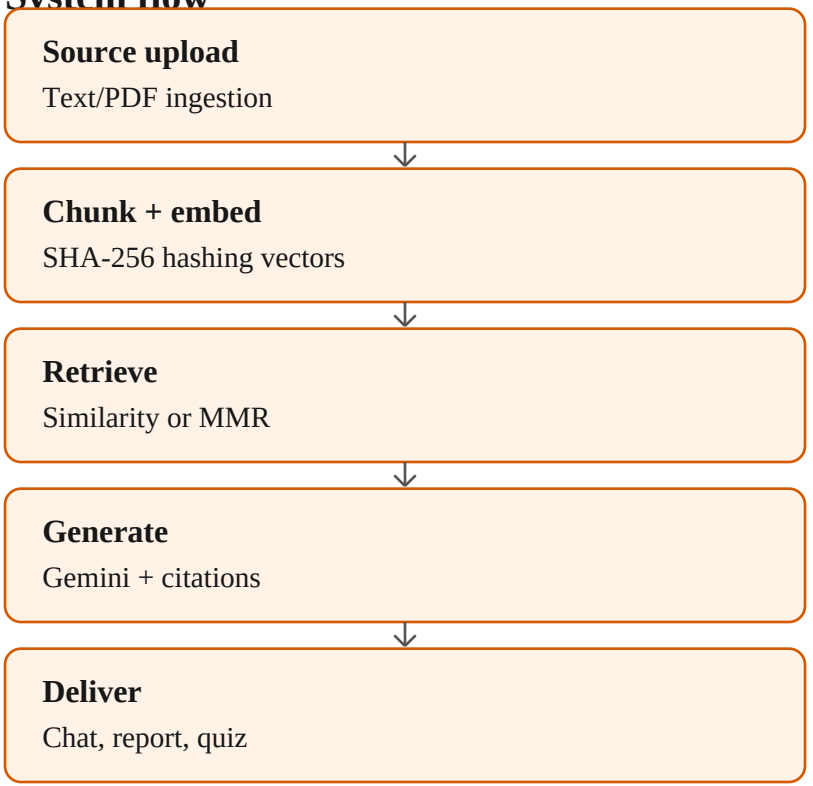
Kennedi Jones-Byrd | Generative AI | October 2026

Project outcome

A deployed, notebook-based RAG study assistant that accepts sources, retrieves relevant evidence, produces cited Gemini answers, creates study artifacts, and compares retrieval strategies.

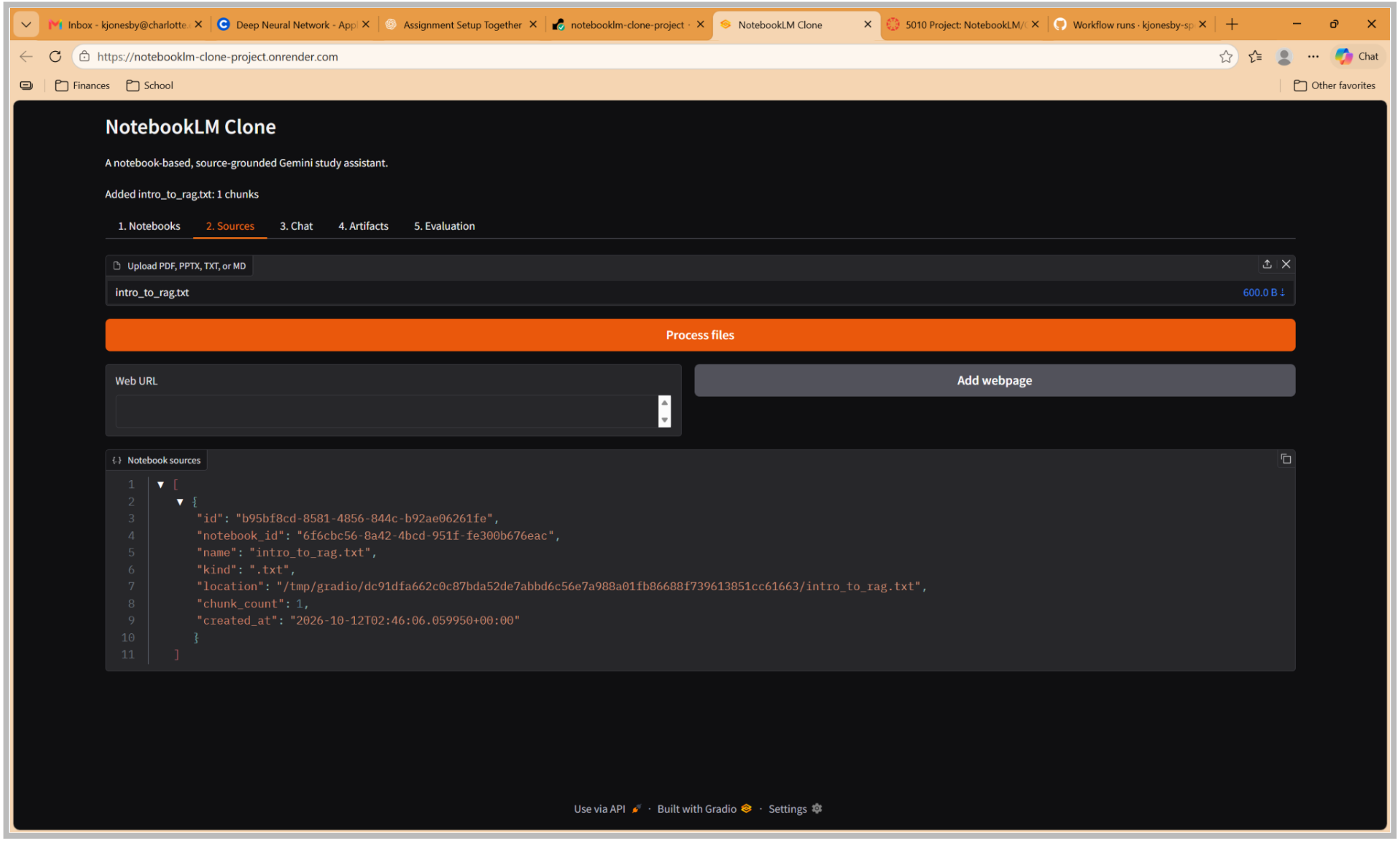
- Created and selected persistent notebooks.
- Uploaded and chunked source documents.
- Stored lightweight vector representations in SQLite.
- Supported similarity and MMR retrieval.
- Generated source-grounded answers with citations.
- Generated report and quiz artifacts.
- Added automated tests, GitHub Actions, Docker, and Render deployment.

System flow



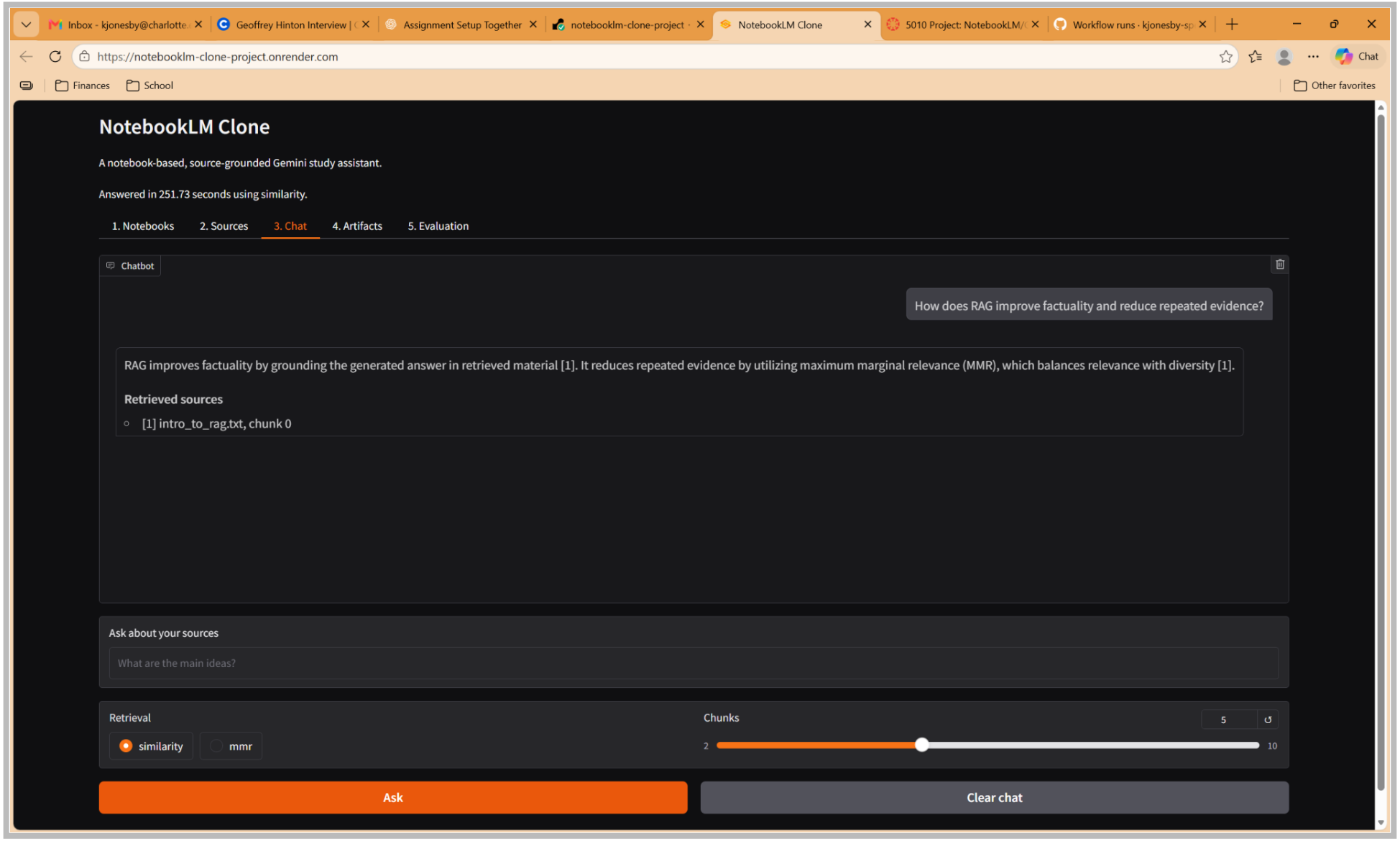
1. Live source ingestion

The deployed application accepted intro_to_rag.txt and stored one source chunk.



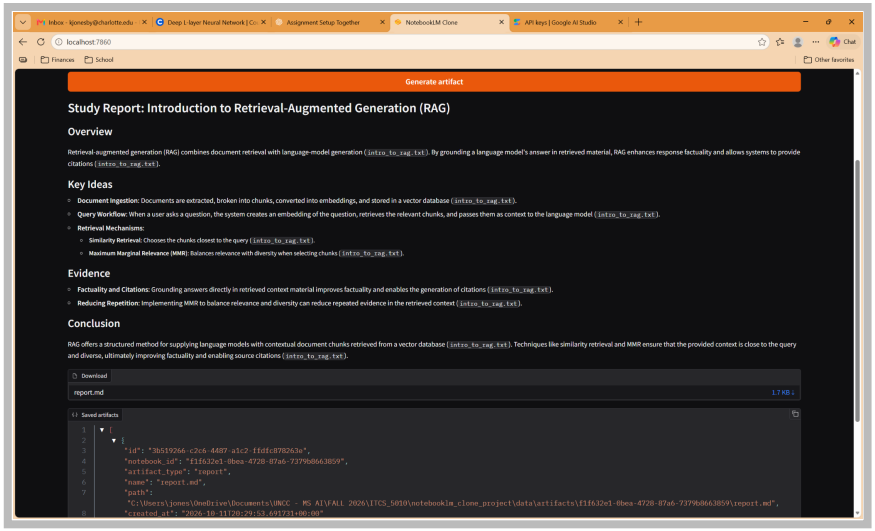
2. Deployed RAG chat with citation

The final Render deployment retrieved the uploaded source and produced a cited answer.

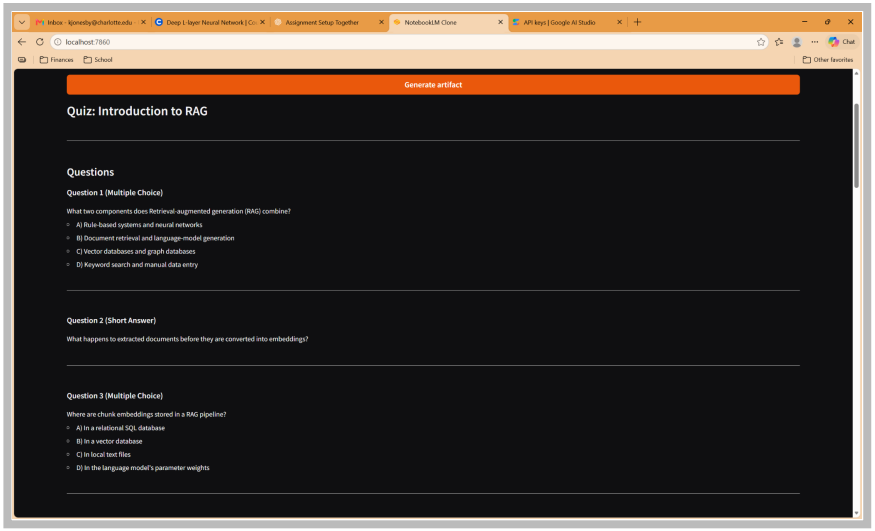


3. Generated study artifacts

The application produces both a source-grounded report and a quiz.



Report generation

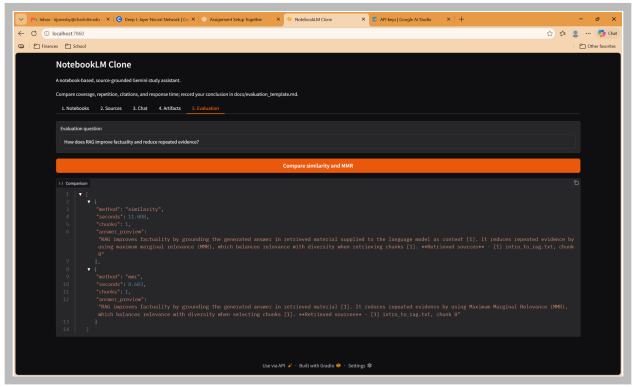


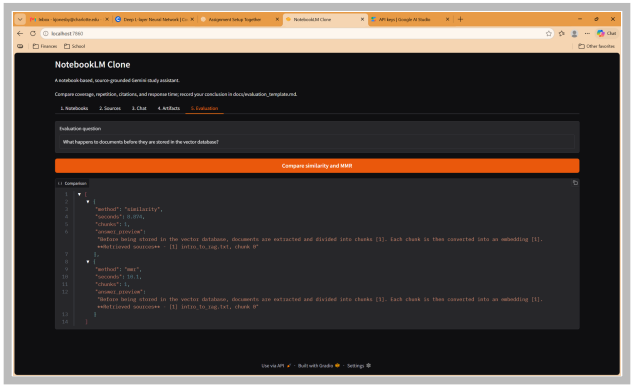
Quiz generation

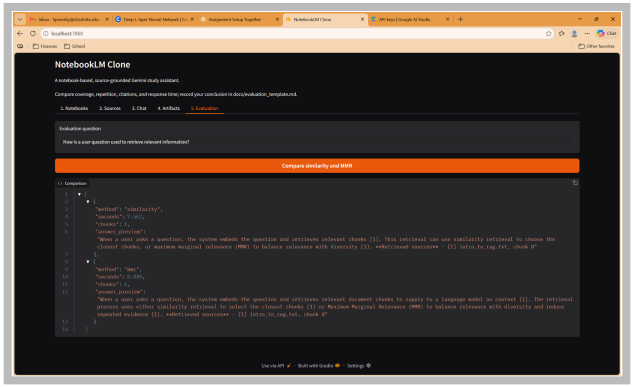
- Artifacts are generated from the selected notebook's stored source material.
- This demonstrates the required study-assistant workflow beyond ordinary chat.

4. Retrieval evaluation

Similarity and MMR were measured across representative RAG questions.



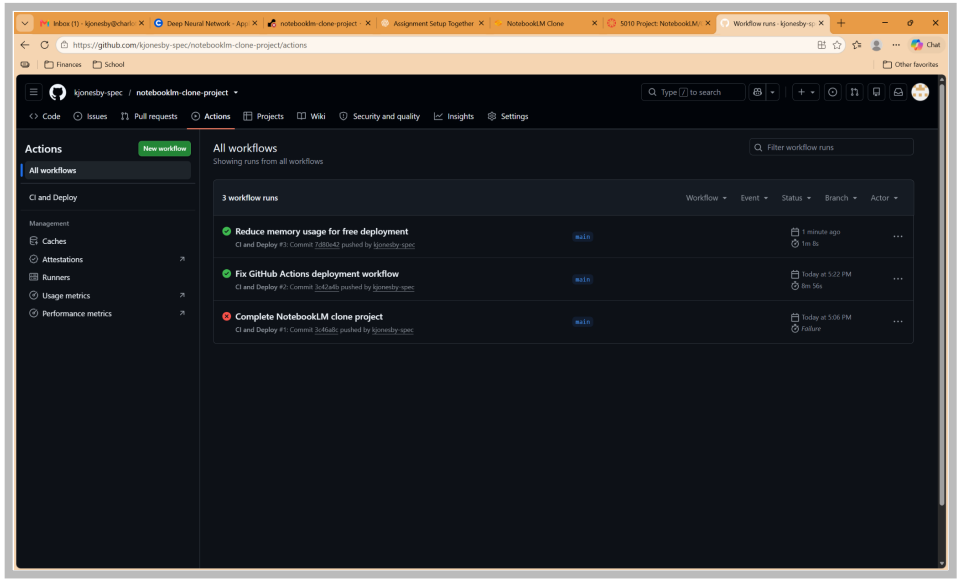




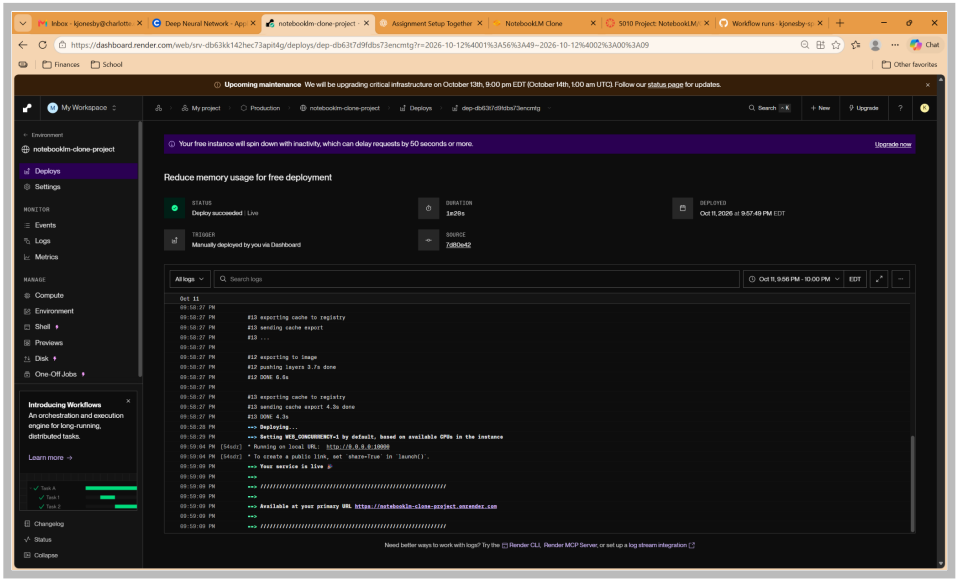
Observed result: both retrieval modes returned the relevant source chunk and produced high-quality grounded answers. With a one-chunk sample source, both methods naturally select the same evidence; MMR becomes more useful as the source collection grows and contains overlapping passages.

5. Automated validation and deployment

The repository tests passed and the Render service reached Live status.



GitHub Actions / automated tests



Render production deployment

- Public repository: https://github.com/kjonesby-spec/notebooklm-clone-project
- Live application: https://notebooklm-clone-project.onrender.com
- The final live chat test confirms the corrected vector field and updated Gemini model configuration.

6. Final submission handoff

Completed technical work and the remaining account-only submission actions.

Completed

- Notebook creation and persistent storage
- Source upload, chunking, embedding, and retrieval
- Cited Gemini chat using the deployed application
- Similarity and MMR retrieval implementations
- Report and quiz artifact generation
- Evaluation evidence, tests, CI/CD, Docker, and Render deployment
- GitHub repository and live application URLs

Student account actions

- Record a 1-2 minute demonstration using Win+Alt+R.
- Upload the video and copy its shareable URL.
- Add the video URL to README.md.
- Confirm the latest GitHub Actions run is green.
- Submit the repository, live app, video, and required report/evaluation items in Canvas.

Suggested demonstration order

Open the live URL -> select the notebook -> show the uploaded source -> ask one question -> point to the citation -> open Artifacts -> show report and quiz -> briefly show the green GitHub Actions run.